

Functional Annotation Report: IscS Cysteine Desulfurase (Q88PK8, PP_0842) in *Pseudomonas putida* KT2440

Gene/Protein Identity Verification

The target is **iscS** (UniProt **Q88PK8**; OrderedLocusName **PP_0842**) from *Pseudomonas putida* strain KT2440, annotated as **Cysteine desulfurase IscS**, EC 2.8.1.7, a member of the **class-V pyridoxal-5'-phosphate (PLP)-dependent aminotransferase family** with the diagnostic domains **Aminotrans_V** (IPR000192/IPR020578), **Cys_deSase_IscS** (**IPR010240**), and the PLP-dependent transferase fold (IPR015424).

Verification outcome — unambiguous. The gene symbol *iscS*, the EC number (2.8.1.7), the protein family, and the domain architecture are fully self-consistent and match the well-characterized bacterial ISC-type cysteine desulfurase. The *P. putida* protein is a close ortholog of the intensively studied *Escherichia coli* IscS (and ~60% identical to human NFS1), so mechanistic and structural conclusions from those orthologs transfer directly. No competing/ambiguous gene assignment was encountered. Where direct *P. putida* biochemical papers are scarce, function is inferred from highly conserved orthologs and from conserved sequence/structure features, and this is flagged below.

1. Summary (Answer to the Research Question)

IscS is a **homodimeric, PLP-dependent L-cysteine desulfurase (EC 2.8.1.7)** that catalyzes the desulfuration of **L-cysteine to L-alanine**, capturing the released sulfane sulfur as an **enzyme-bound cysteinyl persulfide** on its active-site cysteine. It is the **master sulfur-mobilizing enzyme** of the cell: it delivers this persulfide sulfur to the scaffold protein IscU for **iron-sulfur (Fe-S) cluster assembly** (the ISC pathway) and to additional acceptor proteins that channel sulfur into **thiolated tRNA nucleosides, thiamine, biotin, lipoic acid, and the molybdenum cofactor**. It performs these reactions in the **bacterial cytoplasm**.

2. Primary Function and Catalyzed Reaction

Reaction: L-cysteine + [acceptor/reductant] → L-alanine + [acceptor]-S (sulfane sulfur), EC 2.8.1.7.

- IscS is a **PLP-dependent enzyme** that catalyzes the **first committed step of sulfur mobilization** in cofactor biosynthesis

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- Mechanistically, L-cysteine forms an external aldimine with the PLP cofactor; **PLP-dependent cleavage of the C–S bond** yields a **cysteinyl persulfide intermediate** on the catalytic cysteine (Cys328 in *E. coli*), which is regarded as "the hallmark step in sulfur mobilization"

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The carbon skeleton is released as **L-alanine**

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- **Substrate specificity:** the physiological substrate is **L-cysteine**, which serves as the universal sulfur donor; the "product" sulfur is transferred to a bound protein acceptor (or, in vitro, to chemical reductants such as DTT), while alanine is released.

3. Downstream Pathways and Molecular Partners

(a) Iron–sulfur cluster biogenesis (primary output). IscS is a **core component of the ISC assembly system**. It transfers persulfide sulfur to the **U-type scaffold protein IscU**, on which **[2Fe–2S] clusters are built before transfer to recipient apoproteins**

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The crystal structure of the **IscS–IscU complex** shows IscU docking at the IscS active site around **Cys328**

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Native mass spectrometry has captured transient iron- and sulfur-based intermediates on IscU

and supports a **concerted, sulfur-initiated** [2Fe–2S] assembly mechanism

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(b) Sulfur-trafficking hub for thio-cofactors. IscS is a **highly conserved master enzyme** that initiates sulfur transfer to a range of acceptors involved in **Fe–S assembly, tRNA modifications, and sulfur-containing cofactor biosynthesis** — including **thiamine, biotin, lipoic acid, and the molybdenum cofactor (Moco)**

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(c) tRNA thiolation. For **wobble 2-thiouridine (mnm5s2U)** formation, **IscS transfers persulfide sulfur to TusA**, whose desulfurase-stimulating binding launches a relay through **TusBCD → TusE → MnmA** onto tRNA

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IscS similarly supplies **ThiI** (4-thiouridine / thiamine) and interacts with rhodanese RhdA

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Distinct acceptors (IscU, TusA, ThiI) **bind one at a time** at overlapping surfaces centered on Cys328; the **conformational plasticity of the long Cys328 loop** is essential for servicing chemically diverse partners

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Regulatory partners such as **CyaY/frataxin and IscX** also bind IscS but are not themselves sulfur acceptors

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3b. Organism-Specific (*P. putida* KT2440) Evidence

Direct annotation of **UniProt Q88PK8 (ISCS_PSEPK)** confirms the ortholog-based conclusions at the level of the target protein itself: - **404 amino acids, ~44.5 kDa**, functions as a **homodimer** that forms a **heterotetramer with IscU**. - **Active site Cys328** forms the "**cysteine persulfide intermediate**" — the same catalytic cysteine position as experimentally validated *E. coli* IscS. - **PLP cofactor** is covalently bound as **N6-(pyridoxal phosphate)lysine at Lys206** (internal Schiff base), with additional PLP-contacting residues at 75–76, 155, 183, 203–205, 243. - **Cys328 additionally ligates a [2Fe–2S] cluster "via persulfide group" as a "ligand shared with IscU,"** directly encoding the sulfur-hand-off to the scaffold. - **Pathway:** iron–sulfur cluster biosynthesis; **subcellular location:** **Cytoplasm** (HAMAP-Rule MF_00331).

Genomic context. PP_0842 (*iscS*) sits in the canonical, co-regulated **ISC operon: PP_0841 iscR** ([2Fe–2S] transcriptional regulator) – **PP_0842 iscS** – **PP_0843 iscU** (scaffold) – **PP_0844 iscA** (A-type carrier) – **PP_0845 hscB** (co-chaperone) – **PP_0846 hscA** (Hsp70 chaperone). This mirrors the *E. coli iscRSUA-hscBA-fdx* operon, in which **IscR provides [2Fe–2S]-responsive feedback regulation**. The divergent upstream gene **PP_0840 cysE** (serine O-acetyltransferase) ties the operon's neighborhood to cysteine (substrate) biosynthesis.

3c. Regulation of the Assembly Reaction

Not all IscS partners are sulfur acceptors. **CyaY (bacterial frataxin)** and **IscX** bind IscS and act as **regulators of assembly rate** rather than sulfur recipients. Molecular-dynamics/NMR studies of the IscS–IscU and ternary IscS–IscU–CyaY complexes show IscU is firmly anchored yet retains pivotal interfacial motion, and that **CyaY hampers the specific IscU–IscS movements and catalytic-loop flipping required for [2Fe–2S] formation**, giving it an inhibitory role (P 27474202).

In other systems frataxin can conversely **stimulate** desulfurase activity in an **iron-dependent** manner (P 26032732),

so frataxin functions as an iron-responsive rheostat coordinating iron and sulfur supply on the scaffold. Downstream, the **HscA/HscB (Hsp70/J-protein) chaperones and ferredoxin (Fdx)** drive ATP-dependent, redox-assisted transfer of the assembled cluster from IscU to recipient apoproteins.

4. Structure and Catalytic Residue

IscS is a **homodimer** adopting the **class-V (fold-type I) PLP aminotransferase fold**. The essential **active-site cysteine on a mobile loop** (Cys328 in *E. coli*) shuttles the persulfide from the PLP-bound substrate to protein acceptors; its loop flexibility underlies the multi-partner relay

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Crystal structures of IscS–IscU and IscS–TusA complexes, and of *M. tuberculosis* IscS, define this architecture

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5. Localization

Bacterial IscS functions in the **cytoplasm**, where the *isc* operon and Fe–S/cofactor assembly machinery reside; this is explicitly annotated for Q88PK8 (**Subcellular location: Cytoplasm**, HAMAP-Rule MF_00331). Conservation is strong: the human ortholog **NFS1 is ~60% identical** to *E. coli* IscS and localizes to mitochondria (with a cytosolic pool), consistent with IscS acting in the bacterial cytosolic compartment

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6. Supported vs. Refuted Hypotheses

- **Supported:** IscS = PLP-dependent L-cysteine desulfurase producing alanine + enzyme persulfide (EC 2.8.1.7). *Strong, direct biochemical/structural evidence in orthologs.*
- **Supported:** Persulfide sulfur is delivered to IscU for [2Fe–2S] assembly and to TusA/ThiI for tRNA/cofactor thiolation. *Strong; complex crystal structures and reconstituted relays.*
- **Supported:** Homodimeric class-V PLP fold with catalytic Cys on a flexible loop; cytoplasmic. *Strong structural/homology evidence.*
- **Not refuted, but organism-specific caveat:** Direct *P. putida* KT2440 IscS enzymology was not located; conclusions rest on close orthologs (*E. coli*, and the conserved domain signature Cys_deSase_IscS in Q88PK8).

7. Limitations and Future Directions

- Evidence is largely from *E. coli*/*Synechocystis*/*M. tuberculosis*/human orthologs; a *P. putida*-specific kinetic and knockout characterization would confirm partner usage in this organism.
- The exact *P. putida* isc operon composition (iscR, iscU, iscA, hscBA, fdx) and IscR-mediated regulation were not experimentally examined here.
- Quantitative substrate/partner specificity (e.g., selenocysteine handling, relative flux to Fe-S vs. tRNA thiolation) in *P. putida* remains to be measured.

Key References

- Black & Dos Santos 2015, *Biochim Biophys Acta*
([P 25447671](#))
— mechanism, hub role.
- Shi et al. 2010, *PLoS Biol*
([P 20404999](#))
— IscS–IscU and IscS–TusA structures, Cys328 loop.
- Lin et al. 2020, *J Am Chem Soc*
([P 32131593](#));
Bennett et al. 2022
([P 36605734](#))
— Fe-S assembly mechanism on IscU.
- Ikeuchi et al. 2006, *Mol Cell*
([P 16387657](#))
— TusABCDE tRNA 2-thiouridine relay.
- Tirupati et al. 2004
([P 15379559](#))
— products (alanine + cysteinyl persulfide).

- Bühning et al. 2017

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— IscS vs. NFS1 conservation/localization.

- Rybníček et al. 2014

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— IscS structure and Fe-S/oxidative-stress role.

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